

Multimodal Sensor Data Fusion and Ensemble Modeling for Human Locomotion Activity Recognition

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ABSTRACT

The primary research objective of this study is to develop an algorithm pipeline for recognizing human locomotion activities using multimodal sensor data from smartphones, while minimizing prediction errors due to data differences between individuals. The multimodal sensor data provided for the 2023 SHL recognition challenge comprises three types of motion data and two types of radio sensor data. Our team, ‘HELP,’ presents an approach that aligns all the multimodal data to derive a form of vector composed of 106 features, and then blends predictions from multiple learning models which are trained using different number of feature vectors. The proposed neural network models, trained solely on data from a specific individual, yield F1 scores of up to 0.8 in recognizing the locomotion activities of other users. Through post-processing operations, including the ensemble of multiple learning models, it is expected to achieve a performance improvement of 10% or greater in terms of F1 score.

CCS CONCEPTS

• Computing methodologies → Supervised learning by classification.

KEYWORDS

Activity recognition, Neural networks, Machine learning, Multimodal sensors, Human locomotion, SHL Dataset

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1 INTRODUCTION

Recognizing human locomotion activities, which involves determining if and how a person is moving, is crucial for understanding human behavior. With the advancement of various sensors and wearable devices, it has become increasingly easier to record data related to human activities, attracting the interest of many researchers. However, collecting natural behavioral data in everyday settings, as opposed to a controlled laboratory environment, still remains a challenge, with only a few studies having successfully done so [2, 7, 9]. In particular, the Sussex-Huawei Locomotion-Transportation (SHL) dataset was collected through multiple smartphones, placed on four different body parts (i.e., bag, hips, torso, and hand), to obtain multimodal sensor data for real-world human locomotion activities over time.

The previous SHL recognition challenges, held four times to date, aimed to classify eight different human locomotion and transportation using only motion sensor data [11] or only radio sensor data [10]. The multimodal sensor data provided for the 2023 SHL recognition challenge comprises three types of motion data and two types of radio sensor data, collected through multiple smartphones by three experiment participants.

There has already been a lot of research showing that sensor data fusion can improve recognition performance [1, 3, 6, 8, 9, 14]. The major benefits of sensor fusion include (1) increasing the robustness of classification models (e.g., against imperfections or variability in some sensor attributes), (2) improving performance through correlation analysis between sensors, and (3) overcoming situations of temporarily missing sensor data. In light of these prior findings, we have come up with an approach to train deep neural networks (DNN) by aligning and fusing all multimodal sensor data. We also attempted to enhance model performance by combining the prediction results from multiple learning models in a post-processing step known as ‘late fusion’ or ‘decision fusion’.

The paper is organized as follows. Section 2 introduces the characteristics of the dataset provided for the 2023 SHL recognition challenge. Section 3 describes how to fuse multimodal sensor data

to derive feature vectors. Section 4 explains the proposed algorithm pipeline. Section 5 presents detailed experiment results and analyzes the performances in terms of F1 score. Finally, a brief conclusion is presented in Section 6.

2 SHL CHALLENGE DATASET

The dataset for SHL Challenge 2023 is divided into three parts: training (59 days), validation (6 days), and testing data (28 days). While the training data was collected by User1, the other data was collected by User2 and User3. That is, the test data was composed of data recordings from two users, which were not included in the training data. Only a small amount of data recordings from these two users was provided in the validation data. The specific body part where the smartphone was placed to collect data recording in the test data was not disclosed, although it was stated that the data was collected from a single specific location on the body.

From a multimodal sensor perspective, the provided dataset contains the raw data from a total of five sensors, as well as label data for annotating the eight locomotion activity class labels. The three types of motion sensors (i.e., accelerometer, gyroscope, magnetometer) are synchronously sampled at 100 Hz (measured every 10ms), while the two types of radio sensors (GPS and location) are asynchronously recorded at a sampling rate of approximately 1 Hz (once per second) and may be irregularly missing.

As shown in Figure 1, there are significant imbalances in the amount of data for the eight locomotion activities (Still=1, Walking=2, Run=3, Bike=4, Car=5, Bus=6, Train=7, Subway=8) in both the training and the validation data. It should be noted that, in the validation data, the amount of data corresponding to the three locomotion activities (i.e., 'Run', 'Bike', 'Bus') is relatively small. Due to this data imbalance, we opted to use the macro average of the F1 score as the performance metric for the learning models in this study.

The boxplots presented in Figure 2 illustrate the temporal continuity of labeled activities. They indicate that transportation-related activities (i.e., 'Car', 'Bus', 'Train', 'Subway') tends to have longer durations than pedestrian movement activities. Notably, the majority of labeled activities persisted for more than 15 seconds, with the exception of a few sub-second cases of 'Still' and 'Walking' in the training data.

3 MULTIMODAL SENSOR DATA FUSION

To integrate motion data and radio sensor data, we synchronized all sensor data samples by aligning them with fixed-width data frames of 5 seconds, using overlapping sliding windows with 50% overlap. This approach enabled the incorporation of 500 samples from each motion sensor and a maximum of 5 samples from each radio sensor within every data frame.

In the domain of human activity recognition (HAR), numerous studies have demonstrated robust generalization performance through the use of learning models trained on hand-crafted features[11]. Drawing upon several prior studies on feature extraction[4, 5, 14], a total of 106 hand-crafted features are extracted from each 5-second data frame, represented as a vector in the form of

$$F_H(l) = [f_{acc}(l), f_{gyr}(l), f_{mag}(l), f_{loc}(l), f_{gps}(l)]_{106 \times 1},$$

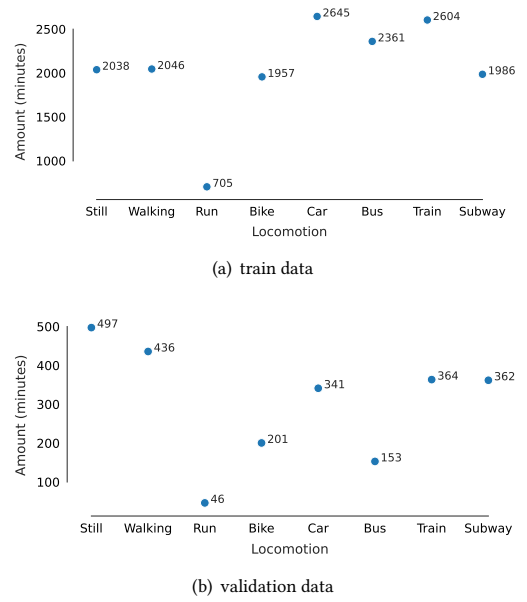


Figure 1: Amount of data per locomotion activity (in minutes)

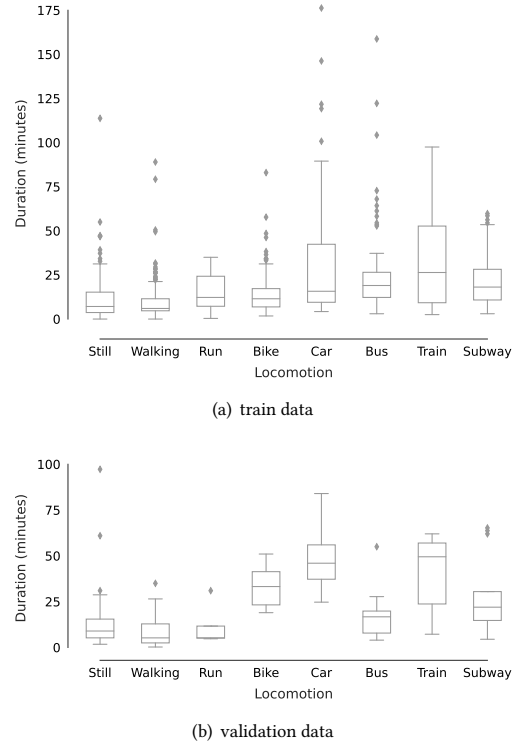


Figure 2: Temporal continuity per locomotion activity

where $f_{acc}(l)$, $f_{gyr}(l)$, $f_{mag}(l)$, $f_{loc}(l)$, and $f_{gps}(l)$ denote a vector of hand-crafted features computed from the l -th frame of the data sequence on the accelerometer (Acc), gyroscope (Gyr), magnetometer (Mag), location, and GPS data, respectively.

$f_{acc}(l)$, $f_{gyr}(l)$, $f_{mag}(l)$: The 3-axis motion sensor data was initially downsampled to 20Hz to remove noise and facilitate subsequent computations. Then, magnitudes were calculated to obtain orientation-independent information, from which 17 features (mean, standard deviation, moments (3rd and 4th), percentiles (25th, 50th, and 75th), value entropy, time entropy, 6 spectral features, and 2 auto-correlation features) were subsequently derived. As addition, 9 statistical attributes were extracted from the 3-axis raw data samples, including mean, standard deviation, and inter-axis correlation coefficients. As a result, each vector comprises 26 features.

$f_{loc}(l)$: We also derived 18 features based on relative location information inferred from the absolute coordinate values, as follows:

- Standard deviation and value change (difference between the last value and the first value) for latitude, longitude, and altitude, respectively.
- Average absolute values and log of range for latitude and longitude, respectively
- Minimum and maximum value of altitude
- Minimum and maximum speed
- Number of updates within each data frame
- Best (lowest) horizontal accuracy
- Diameter(maximum distance between two locations in meters) and its log value

$f_{gps}(l)$: A total of 10 attributes were derived from the GPS sensor data, including maximum, mean, standard deviation, moments (3rd and 4th), percentiles (25th, 50th, and 75th) of signal-to-noise ratio (SNR), average number of visible satellites, Number of changes in the satellite ID of the largest SNR.

$F_H(l)$ was subsequently processed to generate $\bar{F}_H(l)$, representing the result of multimodal sensor data fusion at the l -th data frame (i.e., the 5-second time span). This was accomplished by (1) standardizing $F_H(l)$ using the mean and the standard deviation of all vectors and (2) imputing missing data as zeros. No other data augmentation techniques or external dataset were employed in this study.

4 ALGORITHM PIPELINE

As depicted in Figure 3, which schematizes the proposed pipeline, we utilized a mixture of multiple learning models, incorporating two popular classifiers: four-layer hierarchical neural networks and Random Forest (RF) classifiers. In accordance with the widely accepted notion that a richer amount of data can enhance the performance of a learning model, these classifiers were trained by default using all the provided training data from four distinct body parts.

As previously detailed in the preceding chapter, each feature vector ($\bar{F}_H(l)$) is derived from a 5-second data frame. However, it was observed that locomotion and transportation activities typically persist for a longer periods once initiated, as shown in Figure 2. Therefore, the proposed pipeline adopts a strategy that can accommodate multiple learning models trained to identify the nature of locomotion activities occurring at an extended period (longer than 5 seconds), represented by $(\bar{F}_{H1}, \dots, \bar{F}_{Hk})$. That is, when k consecutive vectors are used as input to train the learning models, it is feasible to analyze time spans of up to $\lceil k/2 \rceil \times 5$ seconds.

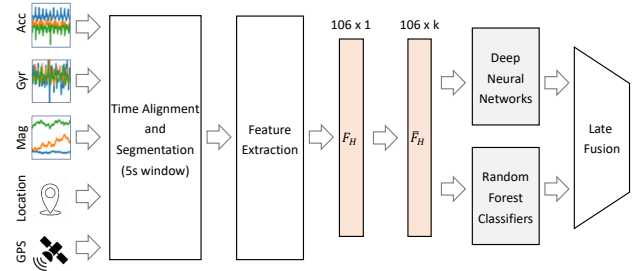


Figure 3: Proposed algorithm pipeline

The DNNs in this study were designed to have identical structural configurations, with the number of layers and hidden nodes remaining constant. Only the input size was adjusted to accommodate the varying number of feature vectors (e.g., $k = \{1, 5, 23\}$). The detailed configurations of the deep learning model are summarized in Table 1. To enhance the generalization capabilities of the hierarchical layers of DNNs, we applied the dropout technique as a regularization method. The final layer of the DNNs is a dense softmax layer, which produces class probabilities for the eight locomotion activities. During the model training phase, the sampling weights for mini-batch sampling were adjusted to be inversely proportional to the proportions of activity class labels, in order to mitigate data imbalance to some extent.

On the other hand, when training a deep hierarchical network model, computational resource consumption increases significantly as the size of input data grows. As such, the DNNs in our analysis were limited to training on inputs with a time span of up to 80 seconds. As an alternative, we were able to analyze time spans of up to 240 seconds using RF classifiers. The hyperparameters of RF models were minimally adjusted around the default values provided by the Scikit-learn library.

During the late fusion phase, the predictions generated by each trained model were combined using a weighted-voting approach as a type of an ensemble strategy. The weights were assigned based on the prediction performance of each model for specific locomotion activities. In particular, our ensemble method combines predictions from models trained on different time steps to produce the final judgment, even if the models were not all trained on the same fixed time step. Furthermore, we fine-tuned the final training models

Table 1: DNN settings

Parameter	Setting value
Input size	(k, 106)
Number of layers	4
Hidden nodes	[96, 48, 32, 8]
Dropout	20%
Optimizer	AdamW
Learning rate(Lr)	0.0002
Lr scheduler	ReduceLROnPlateau
Epochs	Up to 200 (Early stopping)
Batch size	16384 or 4096

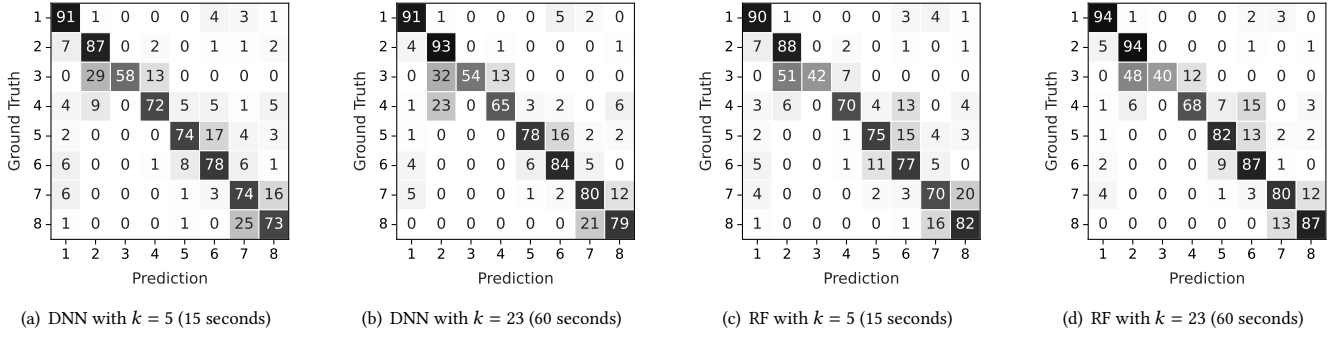


Figure 4: Confusion matrices of the learning models with different number of input size. The eight classes are: 1 - Still; 2 - Walk; 3 - Run; 4 - Bike; 5 - Car; 6 - Bus; 7 - Train; 8 - Subway.

with the validation data before predicting activities in the test data, and applied temporal smoothing[8, 13] to reflect the temporal order of the test data.

5 EXPERIMENTAL RESULTS AND DISCUSSION

All of our experiments are conducted on a PC with a 8-core@3.5GHz CPU, 128GB RAM, and an Nvidia RTX 3090 GPU. The proposed pipeline was implemented using Pytorch and Scikit-learn libraries with the Python programming language.

Table 2 presents the recognition performances, in terms of F1 score, on the validation data using the DNN models trained on the training data. As the input vector size for training DNNs increases, it seems that there is a corresponding increase in F1 scores. When the input vector size exceeded 11 (representing a time span of 30 seconds), increasing the number of training parameters does not improve the model performance. Then, for models with many parameters such as $k = \{23, 31\}$, by adjusting some hyperparameters (e.g., lowering the batch size from 16384 to 4096), we were able to gain slightly higher F1 score values. Thus, for the respective cases of $k = 23$ and $k = 31$, the F1 score values reached 0.804 and 0.816, respectively.

Figure 4(a) and 4(b) display the prediction results from DNNs trained with vector inputs for 15- and 60-second time spans, respectively. It can be observed that the DNN for the 60-second bin, with more vector inputs, exhibits improved classification results for vehicle transportation activities, while the accuracy associated with two other activities ('Run' and 'Bike') is somewhat lower. Figure 4(c)

and 4(d) present the prediction results from RF classifiers trained with vector inputs for 15- and 60-second time spans, respectively. Each RF model achieved F1 scores of 0.76 and 0.80, which are comparable to the performance results of DNNs. On the other hand, by comparing Figure 4(a) and 4(c) (or Figure 4(b) and 4(d)), it is observed that the RF classifiers performed better in identifying the 'Subway' activity.

On the other hand, both DNNs and RFs have difficulty distinguishing between transportation-related activities such as car/bus and train/subway, which is consistent with previous analysis results[10] on the SHL dataset. As shown in the temporal continuity analysis presented in Figure 2, these activities are characterized by their mutual exclusivity in terms of temporal continuity and extended duration. Thus, we trained additional RF models to maximize the classification performance between specific locomotion activities (e.g., Car versus Bus). As presented in Figure 5, the Car-Bus classifier (Figure 5(a)) and the Train-Subway classifier (Figure 5(b)) were trained using 240-second ($k = 95$) time spans to reflect the extended temporal characteristics of transportation-related locomotion activities. This result confirms that the classification results for specific activities could be improved by over 10%, in terms of F1 score, when compared to the previous classification results shown in Figure 4. Further efforts were made to fine-tune the previously trained DNNs using validation data, resulting in an improved classification performance, as shown in Figure 6 (with an F1 score of 0.84). This enabled us to deliberately enhance the classification performance for the activities of User2 and User3, even with only

Table 2: Comparisons of DNN models

k	Seconds	Trainable parameters	F1 score
1	5	17,112	0.7268
3	10	37,464	0.7581
5	15	57,816	0.7727
7	20	78,168	0.7759
11	30	118,872	0.8022
23	60	240,984	0.7952
31	80	322,392	0.7940

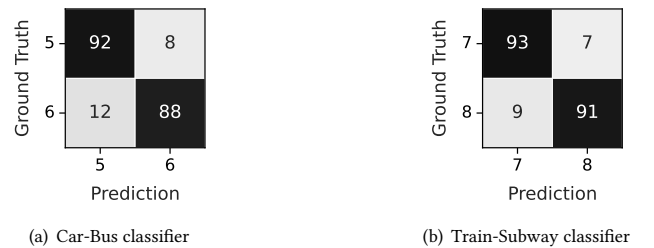


Figure 5: RF classifiers for specific locomotion activities

Ground Truth \ Prediction	1	2	3	4	5	6	7	8
1	93	1	0	0	0	2	2	1
2	8	88	0	1	0	1	0	1
3	0	7	92	1	0	0	0	0
4	4	4	0	89	1	1	0	1
5	2	0	0	0	79	12	3	3
6	7	0	0	1	7	77	6	1
7	6	0	0	0	1	4	71	17
8	2	0	0	0	1	1	15	82

Figure 6: Fine-tuned DNN with $k = 1$ (5 seconds)

5 seconds of time span. Moreover, our findings indicate that these tuning efforts effectively mitigate the difficulties associated with differentiating between pedestrian and bike locomotion activities, a limitation observed in most of the previously trained models (i.e., in Table 2 or Figure 4).

Finally, by employing the late fusion approach for combining the predictions of the learning models trained so far (i.e., DNNs with $k = \{1, 5, 23\}$ as well as RF classifiers for specific activities), while taking into account the temporal continuity of locomotion activities, the algorithm pipeline proposed in this study successfully achieved F1 scores exceeding 0.9 for classifying the eight locomotion activity class labels in the validation data. To summarize the approach used to produce the final prediction result for the test data, for each consecutive time interval, we computed a weighted average of predictions from multiple DNNs and selected the locomotion activity class label with the highest value by default. However, if the top two values corresponded to a particular combination of transportation-related activities, we additionally considered the prediction results of the corresponding RF classifier.

6 CONCLUSION

In this study, we devised an algorithm pipeline to extract features from the multimodal sensor data, and to combine the prediction results of multiple learning models to accurately recognize human locomotion activities. The recognition result for the testing dataset will be presented in the summary paper of the challenge[12].

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